

Fermion Quantum Monte Carlo: Overview

Emilie Huffman



ALF 2.0 Documentation, arXiv:2012.11914
F.F. Assaad and H.G. Evertz, World-line and
determinantal quantum Monte Carlo methods for spins,
phonons and electrons

Outline

QMC Introduction

Hamiltonian

Trotter Decomposition

Hubbard-Stratonovich

Integrating out the Fermions

Configuration Space

Measurements

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Fermionic QMC: Applications

- ▶ Fermionic Quantum Criticality
 - ▶ Gross-Neveu Transitions
 - ▶ Nematic and ferromagnetic critical points in metals
- ▶ Kondo systems: from nano-structures to Kondo lattices
- ▶ Lattice gauge theories coupled to fermionic matter (compact $U(1)$, $\mathbb{Z}_2$)
- ▶ Long-range Coulomb interactions in semi-metals
- ▶ Electron-phonon problems
- ▶ $SU(N)$ quantum antiferromagnets
- ▶ Deconfined quantum critical points
- ▶ Continuum regularization of field theories (Landau level projection)
- ▶ Yukawa-SYK models

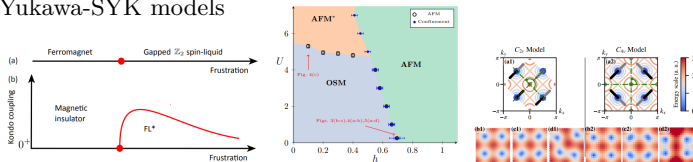
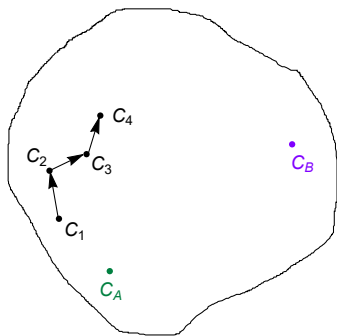


Figure: Left: Hofmann, Assaad, Grover PRB 100, 035118 (2019), Center: Gazit, Assaad, Sachdev, Vishwanath, Wang PNAS 115 (30) (2018), Right: Schwab, Janssen, Sun, Meng, Herbut, Vojta, Assaad, arXiv:2110.02668 (2021)

- Identify configurations C , and weights Ω , such that

$$Z = \text{Tr} \left(e^{-\beta \hat{H}} \right) = \sum_C \Omega(C)$$

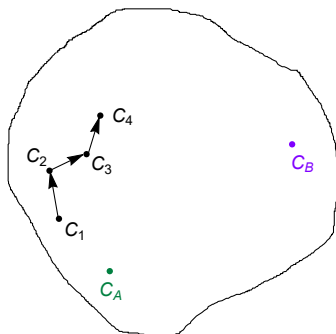


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- We need to compute observables, given by

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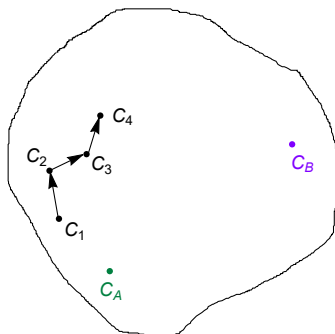


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- **Measurements** $O(C)$ from configurations C with **probability**
 $p = \Omega(C) / \sum_{C'} \Omega(C')$.

Configuration Examples

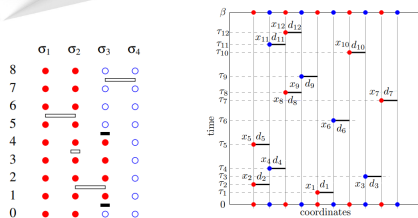


Figure: Left: SSE Lecture notes for Autumn School on Correlated Electrons, Sandvik, Right: Fermion Bags Huffman, Chandrasekharan PRD 2020

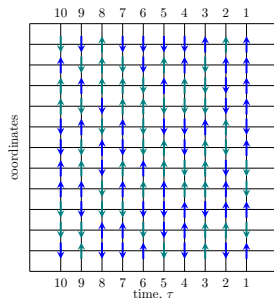
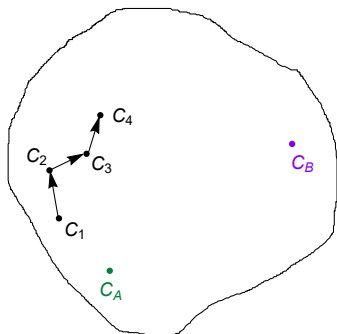


Figure: Auxiliary Field QMC, as implemented in ALF.

Quantum Monte Carlo

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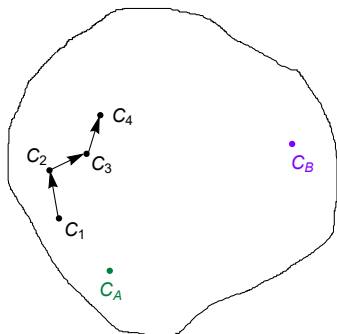


Quantum Monte Carlo

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$$C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow \dots$$



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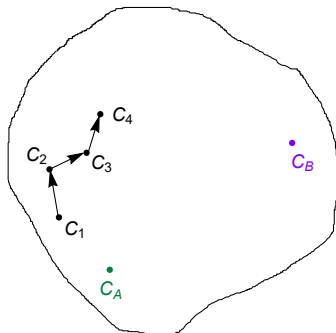
- ▶ **Ergodicity**: one can get from any configuration C_A to any other configuration C_B in the space.

- ▶ **Balance**:

$$\sum_i \Omega(C_i) P(C_i \rightarrow C_k) = \Omega(C_k).$$

- ▶ **Then**:

$$\sum_{k \rightarrow \infty} \sum_i \Omega_0(C_i) P^k(C_i \rightarrow C_j) = a \Omega(C_j)$$



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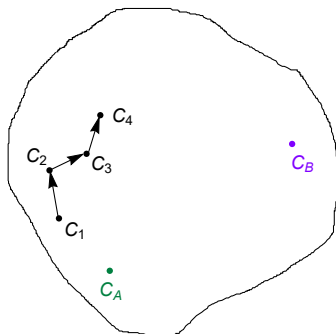
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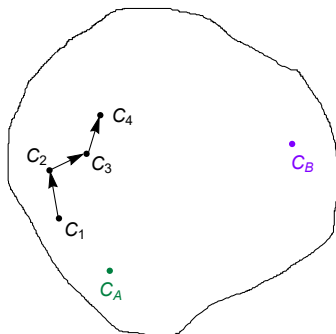
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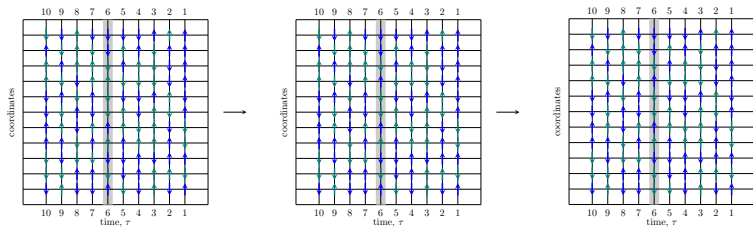


- ▶ Observable from equilibrated configurations:

$$\langle \hat{O} \rangle \approx \frac{1}{n} \sum_{C^*} \hat{O}(C^*)$$

- ▶ Corresponds to central limit theorem, precision $\sim 1/\sqrt{n}$.

Configuration Local Transitions



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AFQMC Hamiltonian Form: Fermionic Terms

- ▶ (Fermionic portions of) eligible Hamiltonians can be written as

$$\hat{H} = \hat{T} + \hat{V},$$

where

$$\hat{T} = \sum_k \hat{c}^\dagger \mathbf{T}^{(k)} \hat{c}, \quad \hat{V} = \sum_k U_k \left(\hat{c}^\dagger \mathbf{v}^{(k)} \hat{c} + \alpha_k \right)^2 \equiv \sum_k \hat{V}_k.$$

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$$\hat{c}^\dagger \mathbf{T}^{(k)} \hat{c} = \sum_{ij} \hat{c}_i^\dagger T_{ij}^{(k)} \hat{c}_j, \quad \hat{c}^\dagger \mathbf{v}^{(k)} \hat{c} = \sum_{ij} \hat{c}_i^\dagger v_{ij}^{(k)} \hat{c}_j$$

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- ▶ Extendable to Hamiltonians that include bosonic (gauge) fields.

Examples of interactions in this form

- ▶ Many interesting interactions can be written in the form

$$\hat{V} = \sum_k U_k \left(\hat{c}^\dagger \mathbf{v}^{(k)} \hat{c} + \alpha_k \right)^2 \equiv \sum_k \hat{V}_k.$$

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- ▶ Hubbard SU(2):

$$\frac{1}{2} (\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow} - 1)^2 = \left(\hat{n}_{i\uparrow} - \frac{1}{2} \right) \left(\hat{n}_{i\downarrow} - \frac{1}{2} \right) + \frac{1}{4}$$

- ▶ Hubbard Mz:

$$-\frac{1}{2} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow})^2 = \left(\hat{n}_{i\uparrow} - \frac{1}{2} \right) \left(\hat{n}_{i\downarrow} - \frac{1}{2} \right) - \frac{1}{4}$$

- ▶ $t - V$:

$$-\frac{1}{2} \left(\hat{c}_i^\dagger \hat{c}_j + \hat{c}_j^\dagger \hat{c}_i \right)^2 = \left(\hat{n}_i - \frac{1}{2} \right) \left(\hat{n}_j - \frac{1}{2} \right) - \frac{1}{4}$$

- Reduce the interacting Hamiltonian to a single particle problem.

$$Z = \text{Tr} \left(e^{-\beta \hat{H}} \right) = \int D\{\Phi(i, \tau)\} e^{-S(\Phi)}.$$

Blankenbecler, Scalapino, Sugar, PRD 24 (1981), 2278,

Hirsch, PRB 31 (1985), 4403

White, Scalapino, Sugar, Loh, Gubernatis, Scalettar, PRB 40 (1989), 506

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Trotter Decomposition

$$\hat{H} = \hat{T} + \hat{V}$$

- Introduce a discrete timestep, $\Delta\tau$, with $\Delta\tau \cdot L_T = \beta$.

$$\begin{aligned} Z &= \text{Tr} \left(e^{-\beta \hat{H}} \right) = \text{Tr} \left(e^{-\beta (\hat{T} + \hat{V})} \right) \\ &= \text{Tr} \left(e^{-\Delta\tau (\hat{T} + \hat{V})} \right)^{L_T} \\ &= \text{Tr} \left(e^{-\Delta\tau \hat{V}} e^{-\Delta\tau \hat{T}} \right)^{L_T} + \mathcal{O}(\Delta\tau^2) \\ &= \text{Tr} \left(\left(\prod_k e^{-\Delta\tau \hat{V}_k} \right) e^{-\Delta\tau \hat{T}} \right)^{L_T} + \mathcal{O}(\Delta\tau^2) \end{aligned}$$

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- **Note 1:** Resulting propagator is not Hermitian
Casas, Murua: *Journal of Mathematical Physics* (2009):

$$e^{-\hat{H}_1} e^{-\hat{H}_2} = e^{-(\hat{H}_3 + i\hat{H}_4)}, \quad \hat{H}_3 = \hat{H}_1 + \hat{H}_2 + \dots, \quad \hat{H}_4 = \frac{i}{2} [\hat{H}_1, \hat{H}_2] + \dots$$

if possible, use symmetric decomposition:

$$e^{-\hat{H}_1/2} e^{-\hat{H}_2} e^{-\hat{H}_1/2}.$$

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- **Note 2:** The systematic Trotter error can be avoided in certain cases.
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- **Note 3:** Higher order methods
Goth: [arXiv:2009.04491](https://arxiv.org/abs/2009.04491)

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$$\hat{H} = \hat{T} + \hat{V}$$

- ▶ $\hat{T}$ is quadratic, and $\hat{V}$ is a sum of perfect squares:

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- Discrete transformation of perfect squares:
(Gauß-Hermite Quadrature, [Goth: arXiv:2009.04491](https://arxiv.org/abs/2009.04491))

$$e^{\Delta\tau \hat{A}^2} = \sum_{l=\pm 1, \pm 2} \gamma(l) e^{\sqrt{\Delta\tau} \Phi(l) \hat{A}} + \mathcal{O}(\Delta\tau^4).$$

$$\begin{aligned} \gamma(\pm 1) &= 1/4 + \sqrt{6}/12, & \gamma(\pm 2) &= 1/4 - \sqrt{6}/12 \\ \Phi(\pm 1) &= \pm \sqrt{2(3 - \sqrt{6})}, & \Phi(\pm 2) &= \pm \sqrt{2(3 + \sqrt{6})} \end{aligned}$$

Hubbard-Stratonovich

$$e^{-\Delta\tau U_k(\hat{c}^\dagger_{\mathbf{v}^{(k)}}\hat{c}+\alpha_k)^2} = \sum_{\{\gamma,\Phi\}} \gamma e^{\sqrt{-\Delta\tau}\Phi\sqrt{U_k}(\hat{c}^\dagger_{\mathbf{v}^{(k)}}\hat{c}+\alpha_k)}$$

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► Interacting problem is reduced to a single particle problem.

- We've written the calculation in terms of single particle problems now.

$$Z = \text{Tr} \left(e^{-\beta \hat{H}} \right) = \int D\{\Phi(i, \tau)\} e^{-S(\Phi)}.$$

Blankenbecler, Scalapino, Sugar, PRD 24 (1981), 2278,

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White, Scalapino, Sugar, Loh, Gubernatis, Scalettar, PRB 40 (1989), 506

Reminder

- ▶ We've written the calculation in terms of single particle problems now.
- ▶ We still need to know how to write it as an action and compute the resulting expression.

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Grassmann Numbers

- $\bar{\xi}_x, \xi_x$, where $x = (i, \tau)$ is a spacetime coordinate.

Algebra

$$\begin{aligned}\bar{\xi}_x \bar{\xi}_y &= -\bar{\xi}_y \bar{\xi}_x, & \xi_x \xi_y &= -\xi_y \xi_x, & \bar{\xi}_x \xi_y &= -\xi_y \bar{\xi}_x, \\ \bar{\xi}_x^2 &= \xi_x^2 = 0,\end{aligned}$$

$$\hat{c}_x \bar{\xi}_y = -\bar{\xi}_y \hat{c}_x, \quad \hat{c}_x \xi_y = -\xi_y \hat{c}_x, \quad \hat{c}_x^\dagger \xi_y = -\xi_y \hat{c}_x^\dagger, \quad \hat{c}_x^\dagger \bar{\xi}_y = -\bar{\xi}_y \hat{c}_x^\dagger$$

Integrals

$$\int \xi_x d\xi_x = 1, \quad \int 1 d\xi_x = 0$$

Coherent State Definition

$$\hat{c}_i |\xi\rangle = \xi_i |\xi\rangle \quad \rightarrow \quad |\xi\rangle = \prod_i \left(1 - \xi_i \hat{c}_i^\dagger\right) |0\rangle$$

$$\langle \bar{\xi} | \hat{c}_i^\dagger = \langle \bar{\xi} | \bar{\xi}_i \quad \rightarrow \quad \langle \bar{\xi} | = \prod_i \langle 0 | \left(1 - \hat{c}_i^\dagger \bar{\xi}_i\right)$$

Identities

$$\langle \bar{\xi} | \xi \rangle = e^{\sum_i \bar{\xi}_i \xi_i}$$

$$\mathbb{1} = \int \prod_i d\bar{\xi}_i d\xi_i e^{-\sum_i \bar{\xi}_i \xi_i} |\xi\rangle \langle \bar{\xi}|$$

$$\text{Tr} \hat{A} = \int \prod_i d\bar{\xi}_i d\xi_i e^{-\sum_i \bar{\xi}_i \xi_i} \langle -\bar{\xi} | \hat{A} | \xi \rangle$$

Coherent State Path Integral

$$\text{Tr} e^{-\beta \hat{H}(\{\Phi\})}$$

$$\propto \int \prod_x d\bar{\xi}_x d\xi_x e^{\sum_{\tau=1}^{L_\tau} [\sum_i -\bar{\xi}_{i,\tau+1}(\xi_{i,\tau+1} - \xi_{i,\tau}) - \Delta\tau \bar{\xi}_{\tau+1} (T + \sum_k v^{(k)}(\Phi_{k,\tau})) \xi_\tau]}$$

$$\equiv \int \prod_\alpha d\bar{\xi}_\alpha d\xi_\alpha e^{-\sum_{\alpha\beta} \bar{\xi}_\alpha M_{\alpha\beta} \xi_\beta} = \det M$$

$$= \det \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & B_{L_\tau} \\ -B_1 & 1 & 0 & \dots & 0 & 0 \\ 0 & -B_2 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -B_{L_\tau-1} & 1 \end{pmatrix}$$

$$B_\tau = 1 - \Delta\tau \left(T + \sum_k v^{(k)}(\Phi_{\tau,k}) \right) = \left(\prod_k e^{-\Delta\tau v^{(k)}(\Phi_{\tau,k})} \right) e^{-\Delta\tau T}$$

Coherent State Path Integral

- Schur's determinant identity:

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det D \det(A - BD^{-1}C).$$

$$\text{Tre}^{-\beta \hat{H}(\{\Phi\})}$$

$$\begin{aligned} &\propto \det \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & B_{L_\tau} \\ -B_1 & 1 & 0 & \dots & 0 & 0 \\ 0 & -B_2 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -B_{L_\tau-1} & 1 \end{pmatrix} \\ &= \det \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & B_{L_\tau} B_{L_\tau-1} \\ -B_1 & 1 & 0 & \dots & 0 & 0 \\ 0 & -B_2 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -B_{L_\tau-2} & 1 \end{pmatrix} \\ &= \dots = \det(1 + B_{L_\tau} \dots B_3 B_2 B_1). \end{aligned}$$

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$$\hat{H} = \hat{T} + \hat{V}, \quad \hat{V} = \sum_k U_k \left(\hat{c}^\dagger_{\mathbf{v}^{(k)}} \hat{c} + \alpha_k \right)^2$$

$$\begin{aligned} Z &= \text{Tr} \left(e^{-\beta \hat{H}} \right) \\ &= \sum_{\{l\}} \left[\prod_{k,\tau} \gamma(l_{k,\tau}) e^{\sqrt{-\Delta\tau} U_k \Phi(l_{k,\tau}) \alpha_k} \right] \det(1 + B_{L\tau} \dots B_3 B_2 B_1), \end{aligned}$$

where

$$B_\tau = \left(\prod_k e^{-\Delta\tau v^{(k)}(\Phi(l_{k,\tau}))} \right) e^{-\Delta\tau T}.$$

Putting it all together and configuration space

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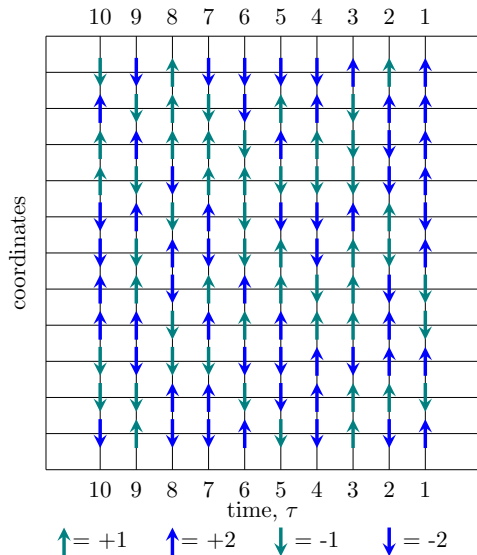
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where

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- Configuration space consists of $C = \{l\} = \{l_1, l_2, \dots, l_{N_\tau \times N_k}\}$, where $l_i \in \{\pm 1, \pm 2\}$. Updates are for these Ising spin degrees of freedom.

Configuration Example



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$$\begin{aligned}\langle \hat{O} \rangle &= \frac{\sum_{\{l\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \hat{O} \right)}{\sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)} = \frac{\sum_{\{l\}} \frac{\text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \hat{O} \right)}{\text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right)} \text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right)}{\sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)} \\ &= \sum_{\{l\}} \langle \langle \hat{O} \rangle \rangle_{\{l\}} \frac{\text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right)}{\sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)}.\end{aligned}$$

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- So we need to select $C = \{l\}$ configurations according to probability $p = \text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right) / \sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)$.

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- We can then compute the observable as

$$\langle \hat{O} \rangle \approx \frac{1}{n} \sum_{\{l^*\}} \langle \langle \hat{O} \rangle \rangle_{\{l^*\}}.$$

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- ▶ So we need to select $C = \{l\}$ configurations according to probability $p = \text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right) / \sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)$.
- ▶ We can then compute the observable as

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- ▶ $\{l^*\}$ are equilibrated configurations, so are randomly selected according to the probability p .